

Hugo Alves Dutra

Software Engineer

Brasília – DF - Brasil

+55 61 982086449

hugo.dutra@hotmail.com

SUMMARY

Full-stack developer with over 15 years of experience. In JavaScript/TypeScript, I worked on the development of scalable web and mobile applications. I have knowledge of microservices architectures, BFF, and distributed systems, with experience in NestJS, ReactJS, Java/Spring, and relational and NoSQL databases. I also worked on systems with CQRS (Command Query Responsibility Segregation) architecture, with write microservices using Oracle, read microservices using MongoDB, and synchronization through corrective messaging. I have experience in modernizing legacy systems and implementing efficient solutions for complex business challenges. I seek to integrate dynamic and collaborative teams to create innovative and impactful solutions.

Skills —

Main languages:

- Javascript
- Typescript
- Java
- C#

Backend:

- NodeJS
- NestJS
- Java Spring Boot

Frontend:

- React
- Angular

Mobile:

- Ionic
- React Native
- Xamarin

Databases:

- PostgreSQL
- Oracle
- MySQL
- SQLite
- Firebird
- MongoDB
- Firebase
- Redis

ORMs:

- TypeORM
- Sequelize
- JPA + Hibernate

EXPERIENCE

Developer Analyst • Fóton Informática • 2025–Present

- I work as a Java Backend Developer at Sicoob, focused on foreign exchange integrations. I develop EJB/JAX-RS services and REST APIs (JSON/XML), standardize contracts, and write technical documentation using Swagger/OpenAPI. I conduct unit tests with JUnit/Mockito, participate in code reviews, and perform deployments to the staging environment via pipelines (Jenkins/GitLab), including configuration, troubleshooting, and monitoring.

Senior Full Stack Engineer • Ília 2022 – 2025

• **Microservices and Microfrontends:** Planning, development, testing, vulnerability remediation, documentation, and implementation of microservices using NestJS (Node) and Java Spring Boot, as well as microfrontends in ReactJS; use of a Design System in Storybook for a robust marketplace platform, with a complete CI/CD pipeline (Jenkins/GitLab), testing and quality analysis with SonarQube, PostgreSQL and MongoDB databases, and monitoring with Grafana and Prometheus. Use of Dynatrace for debugging and problem analysis.

• **Testing:** Implementation of unit and integration tests across all projects.

• **Reference:** Knowledge sharing and best practices to enable fast and effective software deliveries.

Senior Full Stack Engineer • Secretariat of State for Education of the Federal District 2019 – 2022

• Development in ReactJS, React Native, and NodeJS: Applications for managing civil servants/temporary staff and teachers in Brasília.

Cloud:

- Digital Ocean
- Google Cloud Platform

Testing:

- Jest
- Supertest
- React Testing Library

Observability:

- Dynatrace
- Grafana
- NetData

Infrastructure:

- Docker
- Kubernetes
- Istio

CI/CD:

- Github Actions
- Jenkins
- Bitbucket Pipeline
- SonarQube

Agile Methodology:

- Scrum
- Kanban
- Jira

Code Design Principles:

- S.O.L.I.D
- Clean Architecture
- Design Patterns
- DDD
- Microservices

• Maintenance and Migration in Java: Bug fixing and migration of parts of monolithic systems to ReactJS and Node.js (NestJS) projects, using Prometheus and Grafana for monitoring and performance improvement.

• School Platform: Creation of a student enrollment platform, with an Angular frontend and a mobile application built with Ionic, allowing parents real-time access to school information.

• Enrollment Platform: Creation of an enrollment platform for the Volunteer Educator project using ReactJS/TypeScript and Java/Spring. Use of Net Data and Grafana for monitoring.

• Technical Reference: Acting as a technical reference, collaborating with the team and the business area in evaluating the feasibility of new features and adjustments.

Full Stack Pleno Engineer • Infinity 2018 – 2019

• Telehealth System: System for partner clinics: frontend in ReactJS; definition of architecture, routing, authN/authZ, and components; creation of Spring Boot APIs with PostgreSQL and Swagger/OpenAPI.

• Migration – Sales System: Migration of a sales and commissions system (health plans) from Delphi to REST APIs in Java; integration with Firebird and JasperReports; refactoring of the frontend in ReactJS.

Physics Teacher • Secretariat of Education of the Federal District 2010 – 2018

Junior C# Developer • Autotrac Comércio e Telecom. 2008 – 2010

• Windows Service: File encryption service using hardware as a repository for encryption keys.

• Server Monitoring Software: Desktop application that allows the creation of customizable charts for alerts and server metrics monitoring through data capture performed by software called HostMonitor.

Side Project • 2023 – Presente

• AI Customer Service: Development of an API for integration with WhatsApp and web interfaces (widgets) for customer service bots, using NestJS and LangChain. Implementation of support for RAG (Retrieval Augmented Generation), content embedding (PDFs and texts), and generation of synthetic data for fine-tuning AI models. Integration with LLMs for natural language processing and use of OpenAI tools for external tool calls. Storage of conversation context in Redis, conversation history and metrics in MongoDB, and bot parameters, models, temperature, and multidimensional vectors in PostgreSQL with PGVector.

- **AI-Based Ticket Creation:** I implemented an automated support workflow integrating n8n with GLPI (an open-source ITSM platform for asset management, ticketing, and knowledge base) and WhatsApp. Ticket creation—previously handled via phone calls—was moved to a WhatsApp bot, which authenticates users, opens/queries/updates tickets, and retrieves GLPI Wiki articles for contextual responses, reducing friction at first contact.

- In n8n, I orchestrated GLPI webhooks and REST APIs with tools triggered via MCP (Model Context Protocol) to execute on-demand actions (create/update tickets, attach evidence, check status, retrieve KB). The workflows follow the Proxy System support model, with logging, metrics, and exception handling, ensuring automated triage, standardized support, and improved efficiency.

The platform uses MongoDB to store structured support logs and Redis to store per-session context (conversation state and agent variables), ensuring continuity and informed decisions during tool calls. The entire setup runs in containers, enabling portability, scalability, and service isolation.

EDUCATION

Catholic University of Brasília (UCB), Brasília – DF, Brazil — 2002 – 2007

Full Teaching Degree in Physics, with a final graduation project consisting of a 3D simulator of carbon nanotubes developed using Delphi and the OpenGL library.